

CROUP MANAGEMENT GUIDELINES

Croup (laryngotracheobronchitis) is characterized by barking cough, inspiratory stridor and hoarseness of voice or cry
Typical age group 6 months – 3 years old. Peak < 2 years old

Mild Croup	Moderate Croup	Severe Croup
Barking cough, hoarse cry No stridor at rest No chest wall retractions Sats $\geq$ 95%	Little agitation, can be placated Stridor at rest Mild chest wall retractions Sats $\geq$ 95%	Anxious, agitated or pale, tired Stridor decreases with worsening airway obstruction Severe retractions Sats <95%
Oral dexamethasone 0.15mg/kg (max 10mg) OR Nebulized budesonide 2mg if vomiting	Oral dexamethasone 0.6mg/kg (max 10mg) OR Nebulized budesonide 2mg if vomiting Consider nebulised adrenaline early at a dose of 0.5ml/kg of 1:1000 (max 5ml) over 15 mins if the child's clinical status is worsening or there is no response after Dexamethasone	Supplemental oxygen Nebulized adrenaline (can be repeated up to every 15 minutes) Standby for advance airway IV/IM dexamethasone 0.6mg/kg (max 10mg)
Fluids Fever management	Observe for 4 hours after nebulised adrenaline	Admission to PICU/HD Airway team for intubation - controlled setting if possible - avoid neuromuscular blocking

Parental explanation Discharged with follow-up in the next 24 hours or next available	Improvement is usually seen within 30 minutes after giving nebulised adrenaline, but symptoms can recur after 2-4 hours Repeat doses can be given (up to every 15 minutes) if stridor or respiratory distress recurs. Such children should be admitted. Admit to the general paediatric ward for further monitoring of the patient. Manage as for severe croup if deteriorating.	agents - ETT 0.5-1mm smaller than calculated
<i>Relapse is unlikely</i>	<i>5% are expected to return within 24 hours for emergency care when discharged</i>	

Indications for admission

- Children with mild croup can be discharged after one dose of dexamethasone. However, uncertain diagnosis, representation and parental anxiety are indications for admission
- History of severe croup, known structural airway anomaly (subglottic stenosis, haemangioma, Trisomy 21 etc) that predispose to a more severe course
- Medical conditions that pre-dispose to respiratory failure (neuromuscular disorders, chronic lung disease etc)
- Toxic appearance suggesting secondary bacterial infection
- Less than 6 months of age
- Dehydration
- Rebound of moderate croup during 4 hours' observation

Admission disposition

- Respiratory cubicle with contact precautions if not yet tested/tested negative
- Isolated according to etiology accordingly thereafter; please refer to updated Infectious Diseases Guidelines

Indications for admission to PICU/HD

- Respiratory failure requiring intubation
- Persistent severe symptoms requiring frequent nebulized adrenaline
- Underlying conditions predisposing to respiratory failure

Discharge checklist

- ☐ Mild croup with adequate parental understanding of illness and indication to return
- ☐ No respiratory distress (may still have residual stridor)
- ☐ Normal saturations
- ☐ More than 4 hours since Nebulized adrenaline was given and no recurrence of symptoms
- ☐ Fever and fluids management
- ☐ Follow-up available

Medication list

Adrenaline, nebulized	0.5ml/kg of 1:1000 (max 5ml) over 15 mins	Action within 30 min. May be repeated back to back for severe respiratory distress Can rebound within 2-4 hours Cardiac monitoring if repeated dosing
Budesonide, nebulized	2mg neat	More expensive Indicated in vomiting
Dexamethasone, oral, IM or IV	0.15mg/kg mild 0.6mg/kg for mod-severe (max dose of 10mg)	PO route preferred, well absorbed and reaches peak serum concentration as rapid as IM Single dose of corticosteroid is adequate. Repeat doses of corticosteroids are not necessary on a routine basis and may have adverse effects. Moderate to severe symptoms that persist for more than a few days should prompt investigation for other causes of airway obstruction.

References

Bjornson C, Russell K, Vandermeer B et al. Neublized epinephrine for croup in children (Review). Cochrane Database of Systematic Reviews. October 2013

Russell KF, Liang Y, O’Gorman K et al. Glucocorticoids for croup (Review). Cochrane Database of Systematic Reviews. January 2011.

UpToDate Croup: Clinical features, evaluation and diagnosis

UpToDate Croup: Pharmacological and supportive interventions

Canadian Paediatric Society. Acute management of croup in the emergency department. Jan 2017

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